Un ‘deposito vivente’: aperto, relazionale, partecipativo. La trasformazione digitale dei depositi delle opere salvate dal sisma nell’Italia centrale

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Central Italy, and the Apennine belt in particular, are among the Italian territories most exposed to seismic risk, frequently hit by more or less violent events that involve, communities, things, and the environment, at times transforming them irreversibly. The communities that inhabit these mountainous and pre-mountainous areas, difficult to access as they are located outside the major communication routes, have matured over time a seismic culture (Bevilacqua, 1981) that guided them while improving prevention, as well as recovery and restoration of places and things. This seismic culture has evolved in relation to the changing needs and ways of life, but also in relation to the available knowledge and the capabilities related to technical and technological development through history. Cultural heritage, the very rich and articulated set of collective goods, has always been the object of communities' attentions at those junctures, well before the accrual of attention by institutions.

In the long chain of ideas and practices that these communities have developed in response to the vulnerability of the environment, since the second half of the 20th century, repositories have been designed and built as places designated for the sheltering in the emergency phase, and the restoration in the post-emergency phase of cultural heritage works in earthquake-affected areas. These repositories ended up then fulfilling even broader functions, of prolonged shelter, organization and cataloguing often in relation to the impossibility of relocating the works in their original contexts, because badly damaged or destroyed, or even because communities had changed or lost their interest in certain works.

Therefore, repositories have somehow become places where the transformation of the relationship between communities and heritage is processed.

The digital transition has opened new and unimaginable scenarios for these places and for the assets they hold, making them open and therefore potentially alive through the possibility of interaction with the outside world, with communities of reference, with scholars and professionals in the field, and beyond to reach new audiences. Digitization gives access and keys to multidimensional information, data, narratives and perspectives, as well as the possibility to contribute and interact in new ways among disciplines, sectors and stakeholders, organized and unorganized, in the perspective of a co-evolutionary heritage approach (Van Knippenberg et al. 2021).

The contribution we are proposing is aimed at presenting the progress and initial outcomes of a research project aimed at defining a model for digitizing works stored in repositories following the seismic events that struck central Italy between 2009 and 2016. Since then, the repositories where the works were initially provisionally housed have had to fulfil the various functions we have referred to. But widening our gaze into the future, and considering the characteristics of the territory, the depots must in some way already address the issue of prevention and organization in anticipation of the possible need to shelter new goods.

The Project, called DEPOSITS - Digitization and Promotion of Rescued Works in Territories Affected by Natural Disasters[1], aims to address the current needs of repositories and set the direction for the future. The goal is to broaden the perspective of digitization and outline a meaningful strategy both from the point of view of the goods housed in the repositories, considering relocation, lack of

relocation opportunities or completely new placements; and from the point of view of the sheltering contexts in which the goods are stored.

From an asset perspective, the project identifies digital twins (Menaguale 2023) as means that meet the needs of documentation, cataloguing, study, storage, use and enhancement. But not only that. Digital twins can become relational objects (Nicolucci et al. 2022), be situated and contextualized, and be placed at the centre of a circular process in which different communities of reference participate, collaborating: those who contribute to the production and preside over the evolution of digital twins and those who benefit from digital twins for different needs and purposes.

From a context perspective, the project intends to make the repository, itself digitized (H-BIM) (Penjor et al. 2024), a phygital ecosystem, a physical and intangible place, around which different communities organize and act in the perspectives indicated earlier. The physical dimension is associated with the digital one to enable knowledge paths that combine scientific data with history and memory of the past (Benigno 2024), for an enhancement of the tangible and intangible cultural context of heritage. The transformation of the cultural relationship with the works after a disastrous event, and the associated impossibility to rely on a mere relocation, makes it also possible to imagine new forms of use and fruition for the various communities involved. The repository is open and in relationship with the original contexts, with the academic and research community, but also with different and new audiences. The digitized repository is at the same time in relation with the past and, following some of the suggestions of spatial history, becomes the place where one can reconstruct and interpret the relationship with cultural heritage from a historical perspective (White 2010). Through the chosen approach, an attempt is made to build a public heritage that can be transformed over time in relation to emerging needs. Accessibility to the repository is therefore the pivot of this path and, moreover, takes on a particular significance here, in reference to the characteristics of the Apennine territory and its cultural heritage, which is extremely rich, but fragmented, diverse and widespread even in very remote places.

The research intends to hold together the two areas of work, the one dedicated to the goods and the one dedicated to their containers (the repositories), integrating them in order to respond to the specific needs that emerge from the discussion with the different actors involved: the realization of highly specialized technological services; the provision of enabling technologies with respect to contributions (incoming) and fruition (outgoing); the preservation and the possibility of fruition even of the works that will not be able to find a different location; as well as a mediated and concrete accessibility.

The contribution will then present the strategies and methodologies for digitizing the works and repositories of central Italy elaborated by the interdisciplinary and inter-university research group composed of researchers from the four partner universities of the Research Center for Innovation, Digitization, Enhancement and Use of Cultural and Environmental Heritage - CeDiPa: Università degli Studi di Perugia, Università degli Studi dell'Aquila, Università Politecnica delle Marche e Università degli Studi di Teramo. CeDiPa researchers are working with the perspective of making the repositories of goods that were saved from the earthquake "living," open and relational places able to constantly evolve and transform thanks to the circular dialogue with the outside world. A dialogue that founds the digital transformation of heritage and that the project intends to organize from a theoretical and practical point of view. The research thus progresses including the theme of the profound effects of the digital transformation of cultural heritage on the anthropological dimension, with a view to stimulating researchers in the humanities not to limit their sphere of action to the use of technologies, but to make a critical contribution in epistemic terms, in rethinking the relationship between humans and technology. Digital technologies are not intended as mere tools for humanities but as means to endow the humanities with extraordinary transformative power that researchers intend to experiment with in order to combine together the intellectual effort on heritage knowledge, the material effort for conservation actions, and the political effort to accompany the transformation of the relationship of various communities with their heritage.

Notes

1. Funded by " Piano Nazionale di Ripresa e Resilienza – Fondo complementare, Programma unitario di intervento per le aree del terremoto del 2009 e 2016, Misura B, Sub-misura B.4 Centri di ricerca per l'innovazione".

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